



Metabolic differences of the oral microbiome related to dental caries – A pilot study

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ARTICLE INFO

Keywords:

Dental caries
Oral biofilm
Mixed dentition
Microbiome
Metabolome

ABSTRACT

Objective: We aimed to investigate the composition and functions discrepancy of supragingival plaque associated with active deciduous teeth caries in mixed dentitions.

Design: Thirty-three subjects with mixed dentition participated in this study. Children with deciduous teeth caries ($dt \geq 3$) were recruited to the caries group, whereas children without deciduous teeth caries ($dt = 0$) were recruited to the caries-free group. Plaque were collected from deciduous teeth surface and permanent teeth surface respectively. A total of 66 samples of dental plaque were collected and conserved. Illumina 16S rRNA sequencing and diversity analysis were performed for microbiome. Untargeted liquid chromatograph-mass (LC-MS) and partial least squares discriminant analysis were performed for metabolome.

Results: A dominant microbiome of 8 phyla and 22 genera were detected. The alpha diversity indices did not detect differences between the caries and caries-free groups ($p > 0.05$). Beta diversity analysis showed that the microbiota composition was similar between subgroups. Comparative analysis at genus level did not detect difference between caries and caries-free subgroups. The metabolomics analysis yielded 419 biochemical metabolites, 56 of which were related to caries status. Metabolites in glucose metabolism and byproducts of oxidative stress were identified as related to dental caries in mixed dentition. Dominant bacteria are positively correlated with metabolites, such as *Streptococcus* and organic acids.

Conclusions: The upgrade of glucose metabolism and oxidative stress was observed in caries status. Functions discrepancy of oral microbiome may be more pronounced than the composition of oral microbiome with active dental caries in mixed dentitions.

1. Introduction

Dental caries is one of the most prevalent infectious diseases around the world and poses both cost and quality of life burden worldwide. The Global Burden of Disease 2015 Study reported that 573 million children are affected by untreated dental caries in deciduous teeth, while untreated caries in permanent teeth affects 2.5 billion people (Kassebaum et al., 2017). In mainland China, the national survey reported a prevalence of 38.5 % permanent teeth caries in 12-year-old children (Quan et al., 2018), who had just completed the transformation from the primary dentition to the permanent one. The high prevalence indicated that permanent teeth caries may occur and develop in mixed dentition.

It had been observed that a history of deciduous teeth caries is a strong, reliable predictor of future development of permanent teeth

caries (Hall-Scullin et al., 2017). The relevance may be ascribed to physiology, diet, oral hygiene habits and being exposed to similar bacterial populations. With deciduous and permanent teeth coexisting in the oral cavity, the mixed dentition is a special phase where exfoliating and newly erupted teeth can share the same microhabitats. Previous study reported that permanent teeth harbored a higher relative abundance of *Actinobacteria* than deciduous teeth in mixed dentition (Shi et al., 2018). Yang found there was no difference in microbial alpha diversity from supragingival plaque between deciduous teeth and first permanent molar in the same mixed dentition (Yang et al., 2021). The difference of microbiota composition between children with and without caries in mixed dentition is still unclear.

Besides the profile of microbial communities, there remains a paucity of “what is really happening” on the surface of teeth. As dental caries

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<https://doi.org/10.1016/j.archoralbio.2022.105471>

Received 3 February 2022; Received in revised form 31 May 2022; Accepted 2 June 2022

Available online 6 June 2022

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starts from local cariogenic bacteria and their metabolism on carbohydrates, more interest was raised to know about the microecology in teeth biofilm (Takahashi, 2015). In a previous study, metabolites of the glycolytic pathway, pentose phosphate pathway and TCA cycle were detected in supragingival plaque of five adults after glucose rinsing (Takahashi et al., 2010). da Costa Rosa et al. (2021) identified organic acids such as lactate, butyrate, and propionate from the biofilm of 3 teenagers, and found increased levels of organic acids in mature biofilm. Heimisdottir et al. (2021) innovatively applied LC-MS technique to identify metabolites of the supragingival plaque associated with early childhood caries, and found catechin and epicatechin negatively associated with caries status. Still, the studies on comprehensive metabolic features in mixed dentition remained limited.

Mixed dentition is an important stage between the primary and permanent dentition. Children with current active deciduous teeth caries in mixed dentition may have different microbiological traits from caries-free children. This may have a potential influence on the microhabitats of already erupted permanent teeth. Herein, we attempt to elucidate the characteristics of oral bacterial microbiota and metabolites in biofilms collected from mixed dentition with active deciduous teeth caries.

2. Materials and methods

2.1. Ethics approval and subjects recruitment

The study was approved by the Medical Ethics Committee of Hospital of Stomatology, Sun Yat-sen University (approval no.: GHKQ-201910-K4-01). The recruitment is performed under informed consent from the caregivers of participants (aged 7–9 years old) following the World Medical Association Declaration of Helsinki guidelines. All participants were recruited from three public schools in the city of Guangzhou, Guangdong Province, China in May 2021. Guangzhou is located in southern China, with a per capita urban disposable income of \$10,180 annually (National Data, 2019). Exclusion criteria included the history of antibiotic therapy or professional fluoride therapy during the past 3 months, and bleeding on probing of the gingiva. Inclusion criteria included: health in general and all first permanent molars had fully erupted without dental caries.

2.2. Clinical examination

The caries status and oral hygiene of each subject were examined by a clinical dentist (Y.Z.). Before the examination, participants were asked to abstain from tooth brushing in the last 8 hours and avoid eating and drinking for at least 2 hours. The dental examinations were conducted in a dental office, using artificial light from dental chair and Community Periodontal Index (CPI) probe instruments. The caries status was scored by the decayed and filled teeth (dt and ft) index according to the World Health Organization caries diagnostic criteria (World Health Organization, 2013). Children with deciduous teeth caries ($dt \geq 3$) were recruited to the caries group, while those without deciduous teeth caries ($dt = 0$) were recruited to the caries-free group. Permanent teeth in both groups should not present an unmistakable cavity, undermined enamel, or a detectable softened floor or wall (World Health Organization, 2013). The calibration was performed, and the kappa value of inter-examiner was 0.86 in caries examination. Oral hygiene was assessed by Silness & Loe Plaque Index (PLI) (Silness & Loe, 1964). A questionnaire about child's eating habits, brushing habits and family socio-economic background were also collected, which were filled by caregivers.

2.3. Collection of plaque samples

Dental plaque samples were collected after the oral examination. All subjects were asked to rinse their mouths with normal saline for 30 s. In caries group, the supragingival plaque collected from deciduous teeth

sites was named as deciduous caries subgroup (DC) while from permanent teeth sites was named as permanent caries subgroup (PC). In caries-free group, the plaque collected from deciduous teeth sites was named as deciduous caries-free subgroup (DCF) while from permanent teeth sites was named as permanent caries-free subgroup (PCF). The surfaces of teeth were dried by gentle air flow. Cotton was applied to avoid saliva contamination. Biofilm samples were pooled by scraping buccal and lingual teeth surfaces. Collecting samples from cavitated surfaces was avoided. Samples from different subgroup in each subject were collected with different sterile excavators were placed each in a separate tube. Each sample of plaque was then preserved in two sterile Eppendorf tubes (Axygen™, USA), one containing sterile FT (1 ml, for DNA extraction) and the other containing sterile PBS (0.5 ml, for metabolites preservation). Samples were immediately transported to the laboratory on dry ice and stored at -80°C in the laboratory until further processing.

2.4. 16S rRNA amplicon sequencing

The total genomic DNA of the samples was extracted using DNA extraction kit (Magnetic Soil and Stool DNA Kit, TIANGEN, China). The V3–V4 region of bacterial 16S rRNA was amplified with forward primer 338F 5'-ACTCCTACGGGAGGCAGCAG-3' and reverse primer 806R 5'-GGACTACHVGGGTWTCTAAT-3'. Following 2% gel electrophoresis, the PCR products were purified using Qiagen Gel Extraction Kit (Qiagen, Germany).

Sequencing libraries were generated with NEBNext® Ultra™ IIDNA Library Prep Kit (Cat No. E7645) following the manufacturer's recommendations, and the quality was evaluated with Qubit® 2.0 Fluorometer (Thermo Scientific) and Agilent Bioanalyzer 2100 system. Finally, the library was sequenced on the Illumina NovaSeq platform and 250 bp paired-end reads were generated.

Bacterial 16S rRNA gene sequences were clustered into the error-corrected amplicon sequence variants (ASVs) with DADA2 in the QIIME2 (version 2020.06), after which ASVs with abundance less than 5 were filtered out (Bolyen et al., 2019). The taxonomy table was then assigned using QIIME2 classify-sklearn to a pre-trained Silva 138 database (Quast et al., 2013).

The sequencing data are available in the National Center for Biotechnology Information (NCBI), Sequence Read Archive (SRA) database (accession no. PRJNA768619).

2.5. Identification of metabolites by untargeted UHPLC-MS/MS

50 μl of sample pellets were resuspended with 200 μl prechilled 80 % methanol, followed by vortexing and ultrasonication for 6 min. After an ice bath for 5 min, the samples were centrifuged at 15,000 rpm, 4°C for 15 min. The supernatant was freeze-dried at -40°C and then resuspended with 200 μl Bradford reagent and determined the protein concentration using Bradford protein assay method (Bradford, 1976). Normalize the samples to final concentration with LC-MS grade water, based on the protein concentration of corresponding pellets. Finally, the supernatant was analyzed using the UHPLC-MS/MS (ultra-high-performance liquid chromatography and tandem mass spectrometry) system (Dunn et al., 2011).

The analysis was performed with Vanquish UHPLC system (Thermo Fisher, Germany) and a Q Exactive™ HF-X mass spectrometer (Thermo Fisher, Germany). The raw data files generated by UHPLC-MS/MS were processed by Compound Discoverer 3.1 (DC3.1, ThermoFisher) for peak alignment, peak picking, and quantitation for each metabolite. Peak intensities were normalized to the total spectral intensity. Next, peaks were matched with the mzCloud and mzVault database (<https://www.mzcloud.org/>) to obtain accurate qualitative and relative quantitative results. These metabolites were annotated using the HMDB database (Wishart et al., 2018), LIPID MAPS database (Sud et al., 2007) and KEGG database (Kanehisa et al., 2017).

2.6. Bioinformatica and statistical analysis

Demographic and clinical characteristics of the subjects were analyzed using Fisher exact test for categorical variables and Student's t-test for continuous variables. The analysis was performed by SPSS version 26.0 (SPSS Inc., Chicago, IL, USA).

Diversity metrics were calculated and visualized using MicrobiomeAnalyst (Dhariwal et al., 2017) and R (version 3.4.3). The dominant features were identified at a higher relative abundance than 1 %. Alpha diversity was evaluated using Chao1 and Shannon metrics by Kruskal-Wallis (pairwise) test. Before further analysis, features at a minimum count of 4 and 20 % prevalence were filtered to remove low-quality or uninformative features. Beta diversity analysis was performed using Bray-Curtis for distance dissimilarity and was visualized via principal coordinate analysis (PCoA). Significant differences ($p < 0.05$) in beta diversity between groups were tested using permutational multivariate analysis of variance (PERMANOVA) with 999 Monte Carlo permutations. Differentially abundant features were tested with linear discriminant analysis effect size (LEfSe) with a threshold of LDA score > 4 .

For metabolites analysis, statistical analyses were performed using the statistical software R (version 3.4.3), Python (version 2.7.6) and CentOS (version 6.6). Partial least squares discriminant analysis (PLS-DA) was performed to calculate variable importance in the projection (VIP). Univariate t-test was used to calculate the statistical significance (p -value) between caries and caries-free children (DC vs. DCF, PC vs. PCF). The metabolites with VIP > 1 and p -value < 0.05 and fold change (FC) ≥ 1.5 or FC ≤ 0.5 were considered to be differential metabolites. Hierarchical clustering analysis was then applied to speculate the potential relationships among the metabolites. After merging the metabolites identified in positive and negative ionization modes, KEGG pathway analysis was performed to illuminate possible altered metabolic pathways. Hypergeometric test was used to obtain p -value. The KEGG pathway was defined as significantly enriched with the threshold of $p < 0.05$.

Pearson correlation coefficients were calculated to assess for correlations between microbiota and metabolites and p -value < 0.05 was of significant difference.

3. Results

3.1. Demographic and clinical characteristics of the participants

A total of 33 subjects participated in this study, of which 17 subjects met the criteria of the caries group and 16 subjects met the criteria of the caries-free group. The number of decayed teeth (dt) ranged from 3 to 8 and mean dt was 4.53 in caries group. Table 1 lists the demographic and clinical features of the study population. There was no significant difference in age, gender and plaque index between the two groups. There was also no significant difference in food habits, brushing habits and caregivers' educational background between groups.

3.2. Profiling of plaque bacterial microbiome

A total of 66 supragingival biofilm samples were collected and analyzed. After filtering out the low-quality and chimeric tags, a total of 5,442,306 high-quality sequences were obtained, and the average number of each sample was 82,459. After low count flitting, a total of 578 amplicon sequence variants (ASVs) were assigned to 12 phyla, 94 genera.

The dominant microbiome was defined as relative abundance of higher than 1 %. The dominant microbiome was comprised of 8 phyla and 22 genera. Fig. 1A shows the taxonomic distribution of each group at phylum level and Fig. 1B shows the taxonomic distributions of each group at genus level. The 8 most abundant phyla in caries and caries-free group were *Fusobacteriota*, *Proteobacteria*, *Actinobacteriota*, *Bacteroidota*,

Table 1

Demographic and clinical characteristics between caries group and caries free group.

Variables	Caries (n = 17)	Caries-free (n = 16)	p value
Age (month) ^a	101.47 ± 3.59	101.81 ± 3.21	0.775
Gender			0.166 ^b
Female	10	5	
Male	7	11	
decayed deciduous teeth (dt) ^a	4.53	0	
Plaque Index ^a	1.72 ± 0.34	1.59 ± 0.39	0.316
Frequency of sweet intake			0.331 ^b
≥ 1 time a day	4	7	
< 1 time a day ^c	13	9	
Frequency of teeth brushing			0.688 ^b
≥ 2 time per day	12	13	
< 2 time per day ^c	5	3	
Caregiver Help Brushing			0.601 ^b
Yes	1	2	
No ^c	16	14	
Fluoride toothpaste			1.000 ^b
Yes	11	11	
No ^c	6	5	
Caregivers' educational background			0.175 ^b
College and above	12	15	
High School and below ^c	5	1	

^a Represented as mean ± standard deviation, no significance between caries group and caries-free group ($p > 0.05$), by Student's T-Test.

^b No significance between caries group and caries-free group ($p > 0.05$), by Fisher's exact test.

^c Reference group

Firmicutes, *Patescibacteria*, *Campilobacterota* and *Spirochaetota*, accounting for more than 99 % of the total features. *Fusobacteriota* was the top phyla in deciduous teeth subgroup both caries group and caries-free group (DC and DCF) as well as in permanent teeth subgroup in caries-free group (PCF). *Proteobacteria* was the top phyla in permanent teeth subgroup in caries group (PC). The top 5 most abundant genera were *Leptotrichia*, *Fusobacterium*, *Neisseria*, *Corynebacterium*, *Actinomyces*, accounting for 47.2 % of the total. *Streptococcus* accounted for 3.4 % of total genera. *Fusobacterium* was the top genera in deciduous teeth subgroup both caries group and caries-free group (DC and DCF) as well as in permanent teeth subgroup in caries-free group (PCF). *Leptotrichia* was the top phyla in permanent teeth subgroup in caries group (PC).

Alpha diversity analysis was applied to compare the richness and evenness between groups. No significant differences in Chao1 and Shannon indices were found between deciduous teeth subgroups (DC vs. DCF) and between permanent teeth subgroups (PC vs. PCF) (Fig. 2C). Beta diversity analysis revealed no significant separation among the subgroups. The result of PERMANOVA showed no dispersion effect between deciduous teeth subgroups (DC vs. DCF) and between permanent teeth subgroups (PC vs. PCF) (Fig. 2D). Other clinical characteristics cannot explain the shift in microbiome composition either (Table S1). No enriched taxa were identified in LEfSe analysis.

3.3. Profiling of plaque bacterial metabolites

Overall, the metabolomics analysis yielded 419 biochemical metabolites in positive ion mode and 293 in negative ion mode. In deciduous teeth subgroups, 15 upgrade and 13 downgrade metabolites were related to caries trait (Table 2). Clusters of lysophospholipids were of negative association, e.g. lysophosphatidylethanolamine (LPE) and lysophosphatidylglycerol (LPG). Organic acid showed positive relations with caries status, including phosphoenolpyruvic acid and 3-phosphoglyceric acid. In permanent teeth subgroups, 19 upgrade and 9 downgrade metabolites were related to caries trait (Table 2). Very-long-chain fatty acids and oleic acid derivant displayed a negative relation, e.g. 13Z,16Z-docosadienoic acid and oleoyl ethanolamide. Important metabolites in glucose metabolism were of upregulations, e.g.

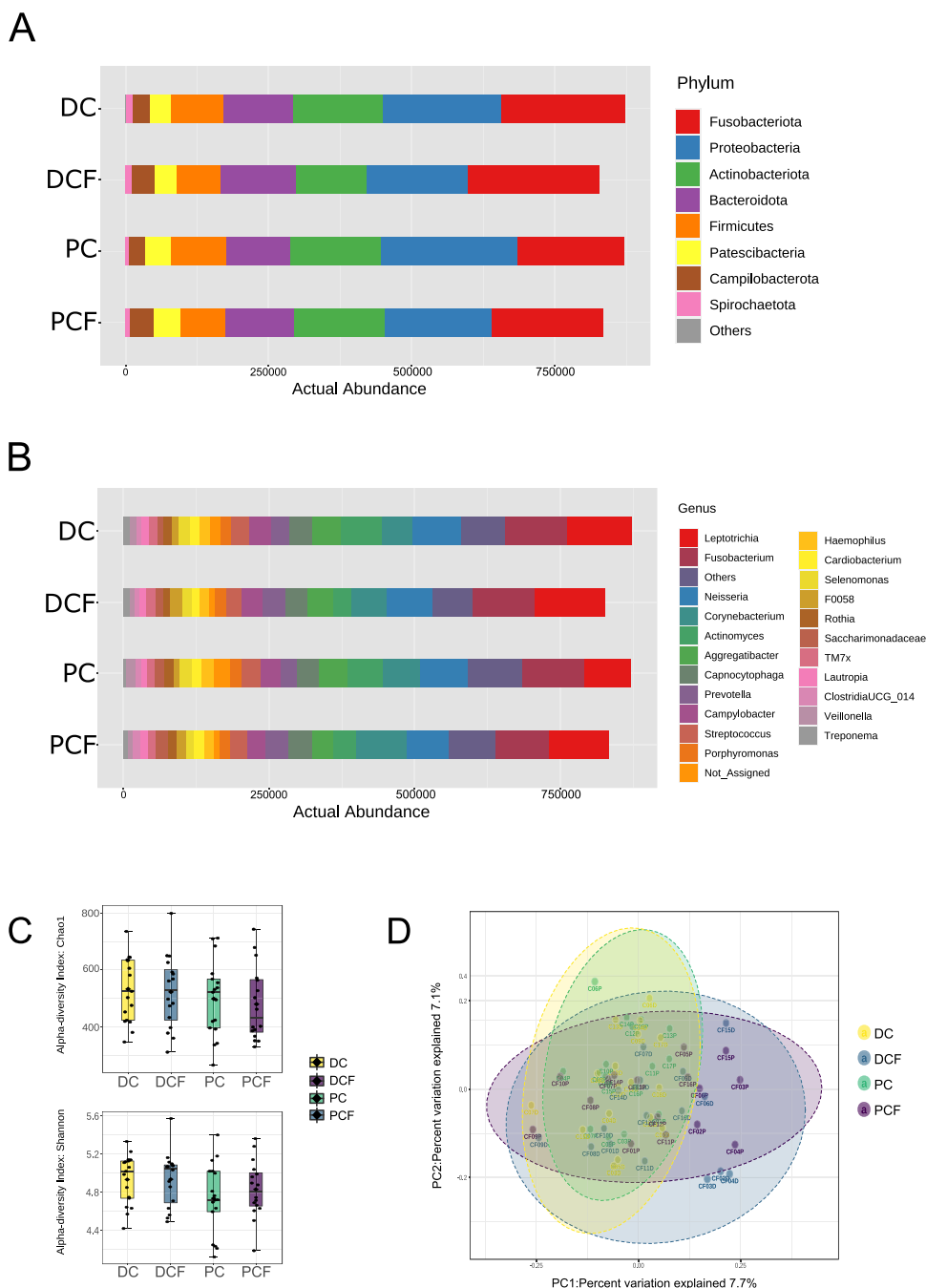


Fig. 1. Microbiome profile and diversity analysis among four groups. DC represents deciduous teeth subgroup in caries group while DCF represents deciduous teeth subgroup in caries-free group. PC represents permanent teeth subgroup in caries group and PCF represents permanent teeth subgroup in caries-free group. (A) Bar graph shows actual abundance of four groups at phyla level. (B) Bar graph shows actual abundance of four groups at genera level. (C) Comparison of the alpha diversity of plaque from each group. Chao1 represents the richness of community, and Shannon represents the community diversity. (D) Principal co-ordinates analysis (PCoA) based on Bray-Curtis dissimilarities of plaque microbiota among four groups.

dihydroxyacetone phosphate and 6-Phosphogluconic acid.

Eight metabolites were related to caries trait in both deciduous subgroup and permanent subgroup (Table 2). Four of them were organic acid (3-phospho-D-glyceric acid, phosphoenolpyruvic acid, 3-Phosphoglyceric acid and L-(+)-Tartaric acid). 8-iso prostaglandin F2 α ethanolamide was notably 52-fold increase in deciduous subgroup and 22-fold increase in permanent subgroup. Hierarchical clustering analysis classified these annotated metabolites into clusters. An increased level of carbohydrates and carbohydrate conjugates metabolites in caries trait was observed, e.g. 3-Phospho-D-glyceric acid (Fig. 2A) and 6-Phosphogluconic acid (Fig. 2B).

Pathway enrichment analysis was applied to find out the most significant differences (Table S2). Six up-regulated signal pathways, including phosphonate and phosphate metabolism, glycolysis/gluconeogenesis, and TCA cycle, were found in the comparison between

caries and caries-free deciduous teeth. Similarly, glycolysis/gluconeogenesis and phosphonate and phosphinate metabolism pathways was found enriched in permanent caries group, compared to permanent caries-free subgroup.

3.4. Correlation of plaque bacterial microbiome and metabolome

To explore the relationships between metabolites and microbiota, heatmaps were generated using Pearson correlation coefficient. Fig. 3A shows correlation between 22 dominant genera and metabolic changes in deciduous teeth subgroups. The contents of LPG and LPS were associated positively with *Leptotrichia*. 8-iso Prostaglandin F2 α Ethanolamide and Oxymatrine were positively associated with *Neisseria* and *Porphyrromonas*. Fig. 3B shows correlation between dominant genera and metabolites altered in permanent teeth subgroup. *Streptococcus*,

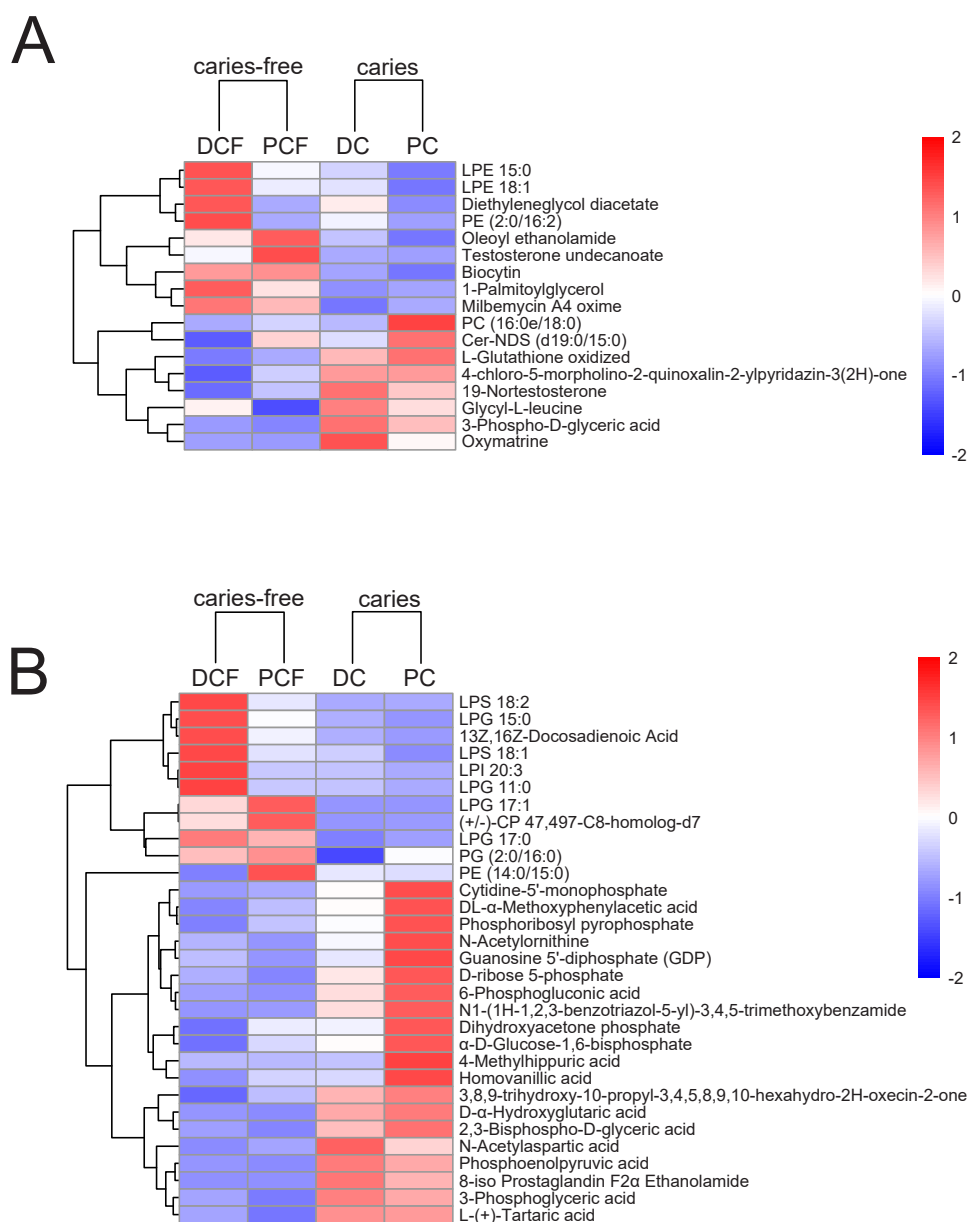


Fig. 2. Heatmap of hierarchical clustering analysis. DC represents deciduous teeth subgroup in caries group while DCF represents deciduous teeth subgroup in caries-free group. PC represents permanent teeth subgroup in caries group and PCF represents permanent teeth subgroup in caries-free group. (A) Metabolites associated with caries status in positive ion mode; (B) Metabolites associated with caries status in negative ion mode.

Haemophilus and *Veillonella* correlated positively with organic acid. *Actinomyces* had a positive correlation with Cytidine-5'-monophosphate and organic acid.

4. Discussion

Multi-omics practices facilitated our understanding of the microenvironment in the oral cavity. The current study set out to explore possible microbial and metabolic differences in mixed dentition with and without deciduous teeth caries by means of multi-omics analysis.

In the current study, the category of dominant phyla and genus are broadly the same as those previously reported microbiome studies in mixed dentition, but the proportions were different (Chen et al., 2020; Xu et al., 2018; Yang et al., 2021). In this study, *Fusobacteriota* was the top phyla of all samples. By comparison, many studies found *Firmicutes* account for the largest fraction of phyla (Crielaard et al., 2011; Chen et al., 2020; Qudeimat et al., 2021). Al-Hebshi et al. reported

Actinobacteria while Xu et al. reported *Bacteroidetes* to be the most abundant taxa (Al-Hebshi et al., 2018; Xu et al., 2018). *Streptococcus* was the most reported genus of the highest proportion (Chen et al., 2020; Crielaard et al., 2011; Qudeimat et al., 2021). However, *Streptococcus* accounted for 3.4 % of total genera in present study and *Leptotrichia* was the top genus. The genus *Leptotrichia* was one of the main genera in healthy oral cavities as well as a consistently observed taxon within hedgehog structures of plaque biofilm (Mark Welch et al., 2016). *Leptotrichia* were traditionally considered to be non-pathogenic, but it can ferment carbohydrates and produce organic acids (Eribe & Olsen, 2017). The discrepancy results could be explained by the individual differences, subject selection process, DNA extraction, sequencing technology and bioinformatics analysis methods.

With regard to the microbial analysis, the richness and diversity of the bacterial communities were similar in the caries and caries-free subgroups. This agreed with previous studies (Al-Hebshi et al., 2018; Chen et al., 2020; Yang et al., 2021). The present study did not find any

Table 2

Altered metabolites associated with caries state in deciduous subgroups.

Subgroup ^a	Correlation ^b	Name	FC ^c	log2FC ^d	p value
DC (vs. DCF)	negative	1-Palmitoylglycerol	0.66	-0.60	0.04
		LPE 18:1*	0.62	-0.68	0.01
		LPE 15:0*	0.59	-0.75	0.01
		Milbemycin A4 oxime	0.49	-1.02	0.02
		Diethyleneglycol diacetate	0.61	-0.70	0.02
		PE (2:0/16:2)	0.59	-0.76	0.03
		LPI 20:3	0.47	-1.09	0.02
		LPG 15:0	0.54	-0.88	0.01
		LPG 17:0	0.58	-0.79	0.02
		LPS 18:2	0.57	-0.81	0.02
		LPS 18:1	0.49	-1.03	0.05
		LPG 11:0	0.46	-1.12	0.00
		PG (2:0/16:0)	0.60	-0.74	0.03
		19-Nortestosterone	1.61	0.69	0.05
		3-Phospho-D-glyceric acid*	2.29	1.20	0.01
	positive	PC (16:0e/18:0)	1.91	0.93	0.03
		4-chloro-5-morpholino-2-quinoxalin-2-ylpyridazin-3(2H)-one	2.10	1.07	0.03
		Cer-NDS (d19:0/15:0)	1.72	0.78	0.04
		Oxymatrine	3.40	1.77	0.05
		Cytidine-5'-monophosphate*	2.13	1.09	0.03
		Phosphoenolpyruvic acid*	1.63	0.71	0.03
		4-Methylhippuric acid	1.91	0.94	0.04
		N-Acetylaspartic acid	2.18	1.12	0.03
		3-Phosphoglyceric acid*	1.61	0.68	0.03
		Phosphoribosyl pyrophosphate	1.55	0.63	0.04
		3,8,9-trihydroxy-10-propyl-3,4,5,8,9,10-hexahydro-2H-oxecin-2-one	1.67	0.74	0.01
		8-iso Prostaglandin F2 α Ethanolamide*	52.77	5.72	0.02
		L-(+)-Tartaric acid*	1.58	0.66	0.03
		Biocytin	0.58	-0.78	0.03
		LPE 18:1*	0.64	-0.65	0.02
		LPE 15:0*	0.64	-0.64	0.02
		Oleoyl ethanolamide	0.30	-1.71	0.00
		Testosterone undecanoate	0.27	-1.88	0.02
		13Z,16Z-Docosadienoic Acid	0.18	-2.49	0.05
PC (vs. PCF)	negative	LPG 17:1	0.46	-1.13	0.04
		PE (14:0/15:0)	0.60	-0.74	0.01
		(+/-)-CP 47,497-C8-homolog-d7	0.41	-1.27	0.05
		Glycyl-L-leucine	1.63	0.70	0.05
		3-Phospho-D-glyceric acid*	2.09	1.06	0.01
		L-Glutathione oxidized	2.59	1.37	0.02
		Homovanillic acid	1.51	0.60	0.03
		Cytidine-5'-monophosphate*	3.55	1.83	0.02
		N-Acetylmethionine	1.56	0.64	0.04
		Dihydroxyacetone phosphate	1.67	0.74	0.03
		Phosphoenolpyruvic acid*	1.54	0.63	0.03
		D-ribose 5-phosphate	1.51	0.60	0.02
	positive	6-Phosphogluconic acid	2.23	1.16	0.01
		2,3-Bisphospho-D-glyceric acid	1.80	0.85	0.04
		3-Phosphoglyceric acid*	1.68	0.75	0.02
		D- α -Hydroxyglutaric acid	1.59	0.67	0.01
		L-(+)-Tartaric acid*	1.80	0.85	0.01
		N1-(1H-1,2,3-benzotriazol-5-yl)-3,4,5-trimethoxybenzamide	1.71	0.77	0.04
		α -D-Glucose-1,6-bisphosphate	2.18	1.12	0.04
		8-iso Prostaglandin F2 α Ethanolamide*	22.43	4.49	0.04
		Guanosine 5'-diphosphate (GDP)	1.93	0.95	0.04
		DL- α -Methoxyphenylacetic acid	1.69	0.75	0.05

^a DC represents deciduous teeth subgroup in caries group while DCF represents deciduous teeth subgroup in caries-free group. PC represents permanent teeth subgroup in caries group and PCF represents permanent teeth subgroup in caries-free group.

^b Negative represents a negative correlation between the metabolites and caries state. Positive represents a positive correlation between the metabolites and caries state.

^c FC, fold change.

^d log2FC, the value of log2 (fold change).

* Metabolites showed association with caries state in both deciduous subgroup and permanent subgroup.

taxonomic differences between groups. As present study pooled dental plaque from teeth surfaces, it was not sensitive enough to explore the difference between the groups. It is reported that there were differences in the microbiota of supragingival plaques on the surface of teeth (Richards et al., 2017). Besides, bacterial 16S rRNA sequencing used in the present study did not reach the species level. Al-Hebshi et al. reported that even sister strains of species can showed differential associations with caries (Al-Hebshi et al., 2018). Changes are expected in

species selection in the caries progression. Deep bioinformatic analysis may overcome this limitation. Ribeiro et al. combined 16S rRNA amplicon sequencing with the BLASTN-based search algorithm for species identification and identified 723 species from 46 plaque samples (Ribeiro et al., 2017). Better bioinformatic analysis methods should be considered in future study.

Over the years, there has been a paradigm shift, that is, it may be more meaningful to figure out what is happening in the biofilm

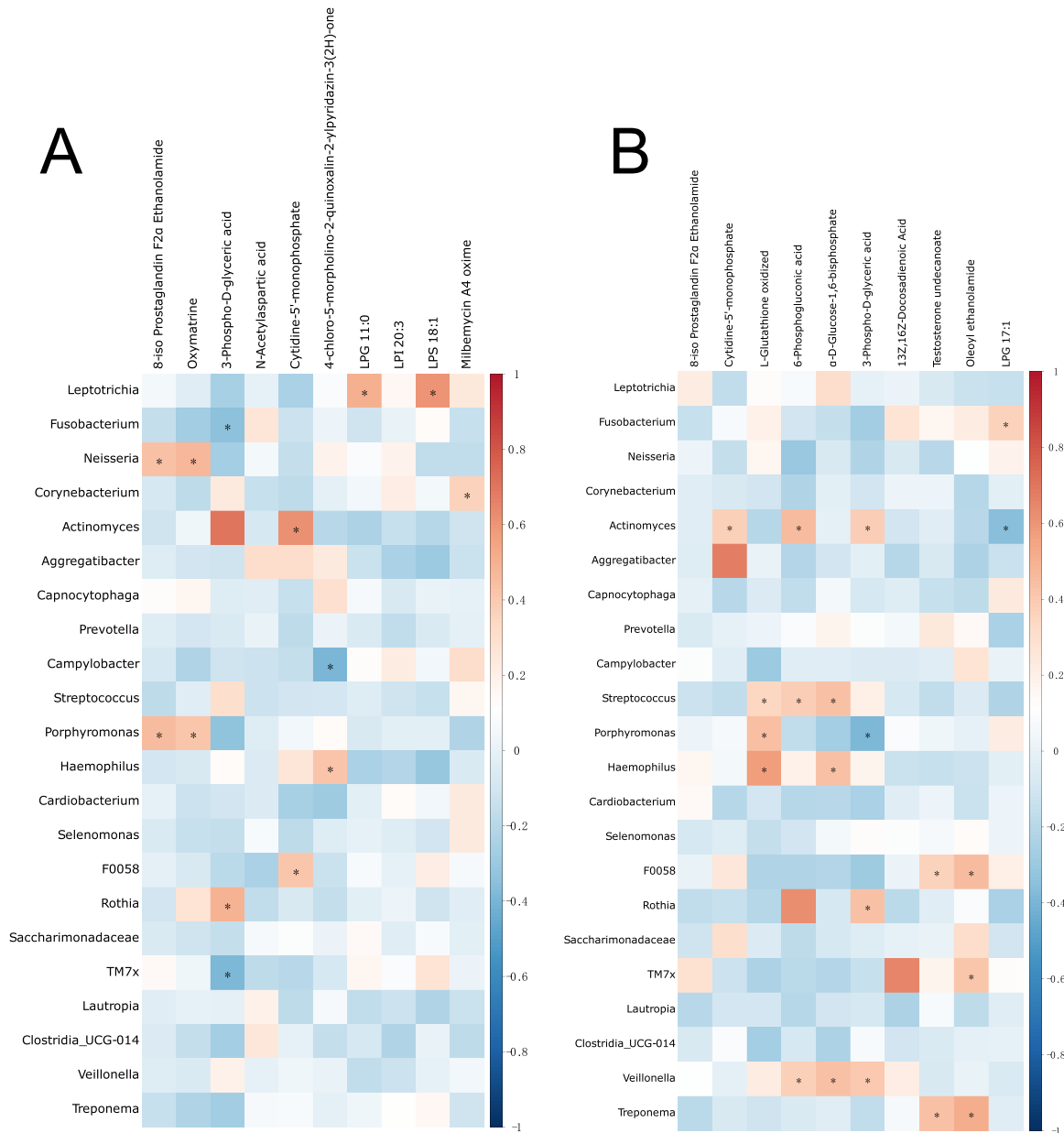


Fig. 3. Heatmap derived from correlation between the identified metabolites and microbiota. DC represents deciduous teeth subgroup in caries group while DCF represents deciduous teeth subgroup in caries-free group. PC represents permanent teeth subgroup in caries group and PCF represents permanent teeth subgroup in caries-free group. (A) Correlation between dominant microbiota and altered metabolites in DC vs.DCF. (B) Correlation between dominant microbiota and altered metabolites in PC vs.PCF.

community, rather than which microbes gain the upper hand (Marsh, 2003). The metabolome is the final step from the genome to metabolic activity, which reflects the ecology of certain microhabitats. Previous studies have disclosed the metabolism pathways in supragingival plaque, including the glycolysis, the pentose-phosphate pathway, and the tricarboxylic acid cycle (TCA cycle) (Takahashi, 2015). In the past decade, great progress has been made in metabolomics technology, and many metabolites have been identified in previous studies. Still, comprehensive simultaneous analysis of multiple metabolites in the same sample has been technically difficult and different analytical platforms are needed to cover the full spectrum of metabolites. The combination of ultra-performance liquid chromatography (UPLC) and tandem mass spectrometry (MS/MS) technique made it possible to identify metabolites in higher specificity and sensitivity, including amino acid, carbohydrate and nucleic acid that may be involved in these pathways or other metabolic activities in the development of dental

caries (Xiao et al., 2012).

In the present studies, we identified a series of altered metabolites that were positively associated with caries states, which can be used to classify caries and caries-free subgroups by hierarchical clustering analysis. The enrichment of metabolic intermediates in glycolysis such as 3-phosphoglyceric acid and phosphoenolpyruvic acid results in lactate production, and therefore leading to the demineralization of dental hard tissue (Takahashi, 2015). The increased level of organic acids and glucose metabolic intermediates can be observed in caries group, including deciduous and permanent teeth sites.

The accumulation of biochemical substances and anti-inflammatory agents related to oxidative stress in dental caries subgroups was also observed. A high multiple change of 8- isoprostaglandin F 2α ethanolamide, which is a lipid peroxidation decomposition product (Alhouayek & Muccioli, 2014; Ayala et al., 2014), was observed in the caries subgroup. In permanent subgroup of caries, a reduction of very-long-chain

fatty acids and oleic acid derivant was observed. These findings revealed that a higher level of lipid degradation and peroxidation may be a characteristic of caries. Most previous studies have shown that the saliva lipid peroxidation levels are elevated in children with dental caries (Subramanyam et al., 2018; Tóthová et al., 2015). The evaluation of oxidative stress metabolites as a risk factor may be a prospective method for dental caries prediction.

Pathway enrichment analysis proved that glycolysis was active in the caries group, though the subjects were restricted to eat at least two hours before sampling. Interestingly, both differential metabolites and pathway analysis concealed a close relationship between caries status and glucose metabolism, even at those teeth sites without dental caries. As it happens, previous studies have shown that the biofilm of children with dental caries may be experiencing active metabolic activities, even without sweet intake at the time (Zandona et al., 2015). The present study also observed enriched glycolysis in caries permanent subgroup, in which the dental plaque samples were from permanent teeth without dental caries. An active level of carbohydrate metabolism in supragingival plaque of individuals with dental caries, even from healthy teeth sites, may raise the tendency of new dental caries in the future.

In the caries deciduous subgroup, caries-related metabolites were positively correlated with *Actinomyces*. It is worth noting that biochemical substances related to oxidative stress are positively correlated with *Neisseria*, which has traditionally been considered as “protecting bacteria” of dental caries (Fakhruddin et al., 2019; Philip et al., 2018). In the caries permanent subgroup, caries-related bacteria *Actinomyces*, *Streptococcus*, *Haemophilus* and *Veillonella* showed a positive correlation with organic acids, consistent with their nature of acid production (Philip et al., 2018). These taxa may play an important role in future dental caries. The finding revealed extensive interlinkage between microbiota and biofilm metabolism. Although microbiome analysis could not classify caries groups and caries-free groups, a shift in the metabolism of caries group was observed, which had a close relationship to certain microbiota. For better comprehensive analysis of microbiota-metabolites, more accurate detection of microbiota is needed, including functional prediction and pathway information. From the clinical point of view, the detection of disease-associated metabolic profiles would be more informative than microbial composition in risk assessment and prevention of dental caries.

The current study is a pilot study. The limitations of this study can be attributed to the small sample size and pooled samples. In addition, the classification of dental caries WHO criteria may include initial caries lesion in caries-free group. The identities of metabolites were based on computational approaches, thus strict experimental verification of the identities of metabolites is needed in future study.

5. Conclusion

The present study drew a preliminary picture of the microbiome and metabolism in supragingival plaque in mixed dentition. The microbiome of deciduous teeth sites in caries and caries-free group seems to be similar, as well as the microbiome of permanent teeth sites. However, metabolites in glucose metabolism and the byproducts of oxidative stress were identified to be related to dental caries in mixed dentition. Some bacteria were positively correlated to organic acid, which indicated that the active acid production of bacteria may be more important than their abundance in dental caries. In conclusion, functions discrepancy of oral microbiome may be more pronounced than the composition of oral microbiome with active dental caries in mixed dentitions.

Funding

This research was funded by Natural Science Foundation of Guangdong Province of China, Grant no. 2021A1515010312” and “Guangdong Financial Fund for High-Caliber Hospital Construction, China, Grant no.

174-2018-XMZC-0001-03-0125/-03”.

CRediT authorship contribution statement

Conceptualization, Yinuo Wang, Yan Zhou and Huancai Lin; Methodology, Yinuo Wang, Yan Zhou and Huancai Lin; Validation, Yan Zhou and Yuwen Zhang; Formal analysis, Yinuo Wang; Investigation, Yan Zhou, Yuwen Zhang, Yinuo Wang and Ting Pan; Resources, Huancai Lin; Data curation, Yuwen Zhang and Ting Pan; Writing – original draft, Yinuo Wang; Writing – review & editing, Yan Zhou, Yuwen Zhang and Huancai Lin; Visualization, Yinuo Wang; Supervision, Huancai Lin; Project administration, Yan Zhou; Funding acquisition, Yan Zhou. All authors have read and agreed to the published version of the manuscript.

Institutional Review Board Statement

The study was conducted according to the guidelines of the Declaration of Helsinki, and approved by the Medical Ethics Committee of Hospital of Stomatology, Sun Yat-sen University (approval no.: GHKQ-201910-K4-01).

Informed Consent Statement

Informed consent was obtained from caregivers of all subjects involved in the study.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability Statement

The sequencing data in this study are available in the National Center for Biotechnology Information (NCBI), Sequence Read Archive (SRA) database (accession no. PRJNA768619). The metabolomic data presented in this study are openly available at <https://www.ebi.ac.uk/metabolights/MTBLS3562>, reference no. MTBLS3562.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.archoralbio.2022.105471](https://doi.org/10.1016/j.archoralbio.2022.105471).

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